

Spring 2025 - A	7	i. 32 ms ii. 2720 ms
	8	i. seq = 6225; ack = 2962 ii. rwnd = 8958 iii. Sf = 2021
Spring 2025 - B	7	i. 40 ms ii. 5200 ms
	8	i. seq = 10522; ack = 5433 ii. rwnd = 7313 iii. Sn = 7358
Fall 2024 - A	6	i. 2275 ms ii. 17500 ms
	7	i. 105 ms ii. 30 ms
	8	ii. seq = 2332; ack = 2013 iii. rwnd = 11468
Fall 2024 - B	6	i. 294 ms ii. 41142.86 ms
	7	i. 155 ms ii. 35 ms
	8	ii. seq = 1887; ack = 2652 iii. rwnd = 9680
Summer 2024 - A	7	i. 37 objects ii. 2.516 seconds
	8	i. seq = 4722; ack = 9274 ii. seq = 4723; ack = 9594 iii. rwnd = 13809

Summer 2024 - B	7	i. 1.35 seconds ii. 3 Mb
	8	i. seq = 2507; ack = 1676 ii. seq = 1676; ack = 2142 iii. rwnd = 6579
Spring 2024 - A	6	N/A -> 173.192.192.0/19 B/A -> 173.192.223.255/19 S/M -> 255.255.224.0
	7	a. 15 objects b. 15 seconds
	8	i. seq = 5563; ack = 1393 ii. ack = 5443 iii. ack = 6323
Spring 2024 - B	6	N/A -> 175.172.0.0/17 B/A -> 175.172.127.255/17 P/M -> 17
	7	a. 8 objects b. 8 seconds
	8	i. seq = 2913; ack = 5294 ii. ack = 2374 iii. ack = 2501
Summer 2023 - A	2(c)	i. 50 ms ii. 1862 ms iii. 4126 ms
	3(b)	Sf = 731 Sn = 770 Starting byte number of window = 731 Last byte number of window = 930
	3(c)	i. seq = 8765; ack = 1026 ii. seq = 1026 iii. ack = 1026

Summer 2023 - B	2(b)	i. 225 ms ii. 6394 ms iii. 9816 ms
	3(b)	Sf = 1300 Sn = 1400 Starting byte number of window = 1300 Last byte number of window = 1799
	3(c)	i. seq = 2231; ack = 5547 ii. ack = 2563 iii. ack = 3018
Spring 2023 - A	2(c)	i. 184 ms ii. 2602 ms (RTT for DNS will be added as we need to fetch the IP address after inserting the url) iii. 1050.3 Mbps
	3(c)	i. seq = 2871; ack = 9861 ii. seq = 11733; ack = 8205 iii. rwnd = 3777
Spring 2023 - B	1(c)	Each file size will be 4 mb (There is a typo in the question) LAN utilization = 15.6% Access Link utilization = 130.1%
	2(c)	i. 88 ms ii. 10056 ms (RTT for DNS will be added as we need to fetch the IP address after inserting the url) iii. 541.3 Mbps
	3(c)	i. seq = 12097; ack = 9425 ii. seq = 10835; ack = 16542 iii. rwnd = 7338

Fall 2022 - A	1(c)	ii. 5.27 s
	2(c)	i. 76 ms ii. 1645 ms or 1540 ms if 5 ms is not included in RTT calculation iii. 4346 ms
	3(b)	urg flag = 1, urgent pointer = 699
	3(c)	i. seq: 2366; ack = 10266 ii. rwnd = 3020 iii. seq = 2755; ack = 10266
Fall 2022 - B	1(d)	1267.6 ms
	2(c)	i. 104 ms ii. 4904 ms or 4784 ms if 4 ms is not included in RTT calculation iii. 11354 ms
	3(c)	i. seq = 11286; ack = 2138 ii. rwnd = 1384 iii. seq = 2138; ack 11286
Summer 2022 - A	1(c)	i. 50 ms ii. 1200 ms iii. 2270 ms
	2(b)	ii. 40
	2(c)	i. seq: 5550; ack: 10236 ii. seq: 11235; rwnd: 7611
	3(c)	i. 10 ii. 1 iii. 1901 iv. 375
Summer 2022 - B	1(c)	i. 220 ms ii. 1540 ms iii. 4385 ms
	2(c)	i. seq: 5550; ack 10236 ii. seq: 11235; rwnd: 8000
	3(c)	i. 11 ii. 0 iii. 1865 iv. 1992

 **There might be errors.**